The Pigeonhole Principle

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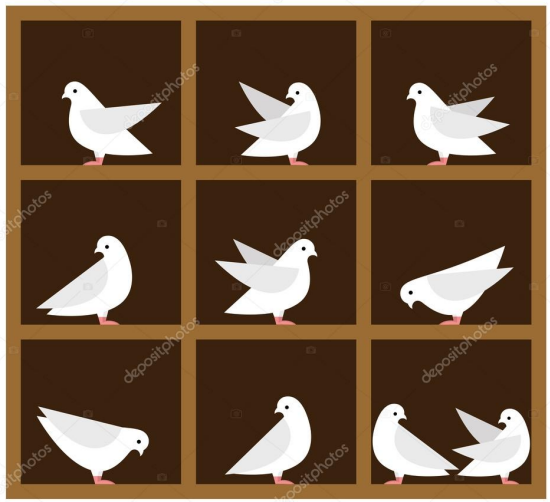
The Pigeonhole Principle: Introduction

Concept

If there are more **objects (pigeons)** than **containers (pigeonholes)**, then at least one container must contain more than one object.

Theorem (Pigeonhole Principle)

If k is a positive integer and $k + 1$ or more objects are placed into k boxes, then at least one box contains two or more objects.



The Pigeonhole Principle: Basic Examples

Example: Birthdays

Among any group of 367 people show at least two share a birthday.

Solution

There are at most 366 possible birthdays (including Feb 29). 367 people (pigeons), 366 days (holes) $\Rightarrow$ collision.

Example: Exam scores

Final exam scores 0–100. How many students guarantee two have same score?

Solution

101 possible scores (0-100), so 102 students guarantee two identical scores.

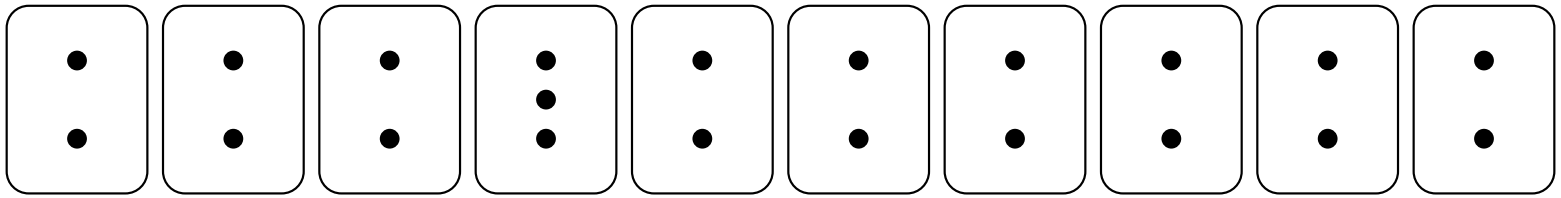
Generalized Pigeonhole Principle

Theorem

If N objects are placed into k boxes, then some box has at least $\lceil \frac{N}{k} \rceil$ objects.

Example

21 students into 10 groups $\Rightarrow$ at least one group has $\lceil 21/10 \rceil = 3$ students.



At least one box has ≥ 3 objects

Useful corollary: To guarantee $\geq r$ objects in some box, need $N = k(r-1)+1$ objects.

Generalized Pigeonhole Principle: Basic Examples

Example: cards (suits)

How many cards must you draw from a 52-card deck to guarantee at least 3 cards of the same suit?

Solution

$k = 4$ suits. Need smallest N with $\lceil N/4 \rceil \geq 3$. So $N = 4 \cdot (3 - 1) + 1 = 9$. (If 8 drawn, could be 2 of each of 4 suit.)

Example: Area codes (Phone numbers)

How many area codes needed so 25,000,000 phones can each have distinct 10-digit numbers (NXX-NXX-XXXX with rules)?

Solution

There are $8,000,000$ distinct NXX–XXXX combinations for the local seven digits. Dividing $25,000,000$ by $8,000,000$ gives $\lceil 25/8 \rceil = 4$. Therefore, at least **4 area codes** are required.

Practice Problem 1: Socks (simple)

Problem: Drawer has 12 brown and 12 black unmatched socks. In dark, how many socks must be taken to be sure of:

1. at least two of the same color?
2. at least two black socks?

Socks (solution)

1. With two colors ($k = 2$), to force ≥ 2 of same color: worst case pick one brown and one black (2 picks) — but that already gives 2 socks possibly different. To guarantee two of same color we need 3 picks by pigeonhole ($\lceil 3/2 \rceil = 2$). So answer: **3 socks**.
2. To guarantee two black socks in worst case you could pick all 12 brown first. So you may need $12 + 2 = 14$ picks to ensure two blacks.

Practice Problem 2: Balls (colored)

Problem: Bowl has 10 red and 10 blue balls. How many must be selected (blindly) to ensure:

1. at least three balls of same color?
2. at least three blue balls?

Balls (solution)

1. Two colors, want ≥ 3 in some color. Smallest N with $\lceil N/2 \rceil \geq 3$ is $N = 5$. So pick **5 balls** guarantee some color occurs ≥ 3.
2. To guarantee 3 blue in worst case draw all 10 red first then 3 blue $\Rightarrow$ 13 picks needed.

Practice Problem 3: Remainders

Problem: Show that among any $d + 1$ integers, two have the same remainder modulo d.

Solution: Remainders modulo d are $0, 1, \ldots, d - 1$ (i.e., $k = d$ boxes). With $d + 1$ integers (pigeons) into d remainders (holes), some remainder repeats. Hence two integers share same remainder.

Practice Problem 4: States & students

Problem: Minimum number of students (each from one of 50 states) to guarantee at least 100 from the same state?

Solution: Let $k = 50$, want $\lceil N/50 \rceil \geq 100$.
Smallest N solving this is $N = 50(100 - 1) + 1 = 50 \cdot 99 + 1 = 4951$.
So **4951 students** guarantee some state supplies ≥ 100 students.
(General formula: $k(r - 1) + 1$.)

Additional exercises

1. Among any 5 integers, show two have same remainder mod 4.
2. How many numbers from $\{1, \ldots, 6\}$ must be chosen to guarantee a pair summing to 7? (Hint: use pigeonhole on complementary pairs.)
3. Show: among $n + 1$ integers $\leq 2n$ there exists one that divides another.